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RESUME - ABSTRACT	**NOM DE LA CAMPAGNE :** **STORM**

RESUME

• Texte synthétique résumant l'ensemble des documents 1 à 6, lisible par un non spécialiste

Entre l'Australie et l'Antarctique, la dorsale Sud-Est Indienne (SEIR), à taux d'ouverture intermédiaire, est surtout connue pour la présence de la Discordance Australie-Antarctique (AAD), une zone à faible approvisionnement magmatique de l'axe d'accrétion. Le projet de campagne STORM (South Tasmania Ocean Ridge and Mantle) s'intéresse à la zone située à l'est de la discordance, où les cartes d'anomalies de gravité déduites de l'altimétrie révèlent la présence de nombreux volcans hors-axe et de chaînes de volcans, entre la Tasmanie et l'Antarctique. Le grand nombre de monts sous-marins suggère un approvisionnement en magma anormalement élevé, non seulement par rapport à la zone de la discordance, faiblement alimentée en magma, mais aussi par rapport aux sections de dorsales intermédiaires habituelles. L'obliquité des rides volcaniques reflète la présence d'un flux asthénosphérique vers l'ouest. Nous souhaitons dans ce projet tester l'hypothèse selon laquelle le volcanisme hors-axe observé résulte de la fusion partielle de manteau de type Pacifique, qui flue du sud-est vers le nord-ouest, sans doute depuis l'ouverture océanique entre la Tasmanie et l'Antarctique. Le flux de manteau à l'origine de l'anomalie de production magmatique pourrait être en partie arrêté par les grandes failles transformantes. Le lien éventuel entre un tel flux de manteau du domaine Pacifique vers l'Indien et le point chaud de Balleny n'est pas évident non plus. L'objectif principal de ce projet est de contraindre la dynamique du manteau à la frontière entre les « boîtes » de convection Pacifique et Indienne.

Nous proposons une campagne en mer pour cartographier et échantillonner le plancher océanique à l'axe de la dorsale Sud-Est Indienne entre la limite est de l'AAD (près de 128°E) et la faille transformante à 140°E, ainsi que les volcans hors-axe identifiés sur les cartes altimétriques (en particulier entre 136°E et 138°E). Nous prévoyons également de lever une carte du système complexe transformant à 140°E (Faille transformante George V). Nous utiliserons le « wax core » pour échantillonner les verres basaltiques de l'axe, et des dragages pour échantillonner les volcans hors-axe et le système de faille transformante. Des mesures in-situ dans la colonne d'eau (néphélométrie et capteurs chimiques Fe/Mn/CH_4 sur CTD wax cores) près de l'axe et dans la faille transformante permettront de mettre en évidence les zones d'activité hydrothermale.

ABSTRACT

Traduction en Anglais du résumé

The intermediate-spreading South-East Indian Ridge (SEIR) between Australia and Antarctica is mostly known for the presence of the Australia-Antarctic Discordance (AAD), an area of low magma budget on the ridge. The STORM (South Tasmania Ocean Ridge and Mantle) project focuses on the area east of the discordance, where gravity highs observed on satellite-derived maps of the flanks of the Southeast Indian Ridge between Tasmania and Antarctica reveal numerous volcanic seamounts and seamount chains. The large number of off-axis seamounts suggests an anomalously high magma supply under the southern flank of the SEIR, not only with respect to the magmatically-starved discordance area, but also with respect to regular sections of intermediate-spreading mid-ocean ridges. The obliquity of the ridges reflects a regional westward asthenospheric flow. We want to test the hypothesis that the observed off-axis volcanism results from the partial melting of Pacific-type mantle as it flows from the southeast to the northwest, a process which likely started with the opening of the oceanic spreading center between Tasmania and Antarctica. The mantle flow at the origin of the magmatic anomaly might be partly dammed by the large transform faults. It is not clear how this regional flow from Pacific to Indian domains relates to the Balleny hotspot. The main objective of this project is to constrain the mantle dynamics at the border between the Pacific and the Indian convecting "boxes".

We propose a cruise to survey and sample the South East Indian Ridge axis between the eastern AAD (near 128°E) and the 140°E transform fault, and the off-axis volcanoes visible on altimetry maps (especially between 136°E and 138°E). We plan to map the 140°E (George V) complex transform fault system. We plan to use wax coring to sample the ridge axis basalts and dredging to sample the off-axis seamounts and the transform fault system. In-situ measurements in the water column (nephelometry, and chemical sensors Fe/Mn/CH_4 on CTD and wax cores) near the axis and in the transform faults will detect the presence of hydrothermal activity.

DOCUMENT N° 1	**NOM DE LA CAMPAGNE :** **STORM**

PROJET SCIENTIFIQUE, TECHNOLOGIQUE OU TECHNIQUE

The intermediate-spreading South-East Indian Ridge (SEIR) located between Australia and Antarctica is mostly known for the presence of the Australia-Antarctic Discordance (AAD), an area of low magma budget on the ridge. The STORM (South Tasmania Ocean Ridge and Mantle) project focuses on the area east of the discordance, where gravity highs observed on satellite-derived maps of the flanks of the Southeast Indian Ridge between Tasmania and Antarctica reveal numerous volcanic seamounts and seamount chains.

1. Scientific rationale – The South East Indian Ridge East of the Australian Antarctic Discordance

The SEIR is bounded to the west by the Rodrigues Triple Junction (RTJ ; 70°E, 25°30'S) and to the east by the Macquarie Triple Junction (MTJ ; 160°E, 65°S) (Figure 1). It was created about 85 m.y. ago (anomaly 34) by the rifting of Gondwana (Veevers et al., 1982). Its spreading rate increased about 50 m.y. ago from ultra-slow to intermediate, after the major plate reorganization in the Pacific Ocean, and has remained stable since then (Royer et al., 1989 ; Whittaker et al., 2007). The SEIR axis has also been migrating in a main NNE-trending direction relative to the fixed hotspot reference frame for the last 43 m.y. (Gripp and Gordon, 1990).

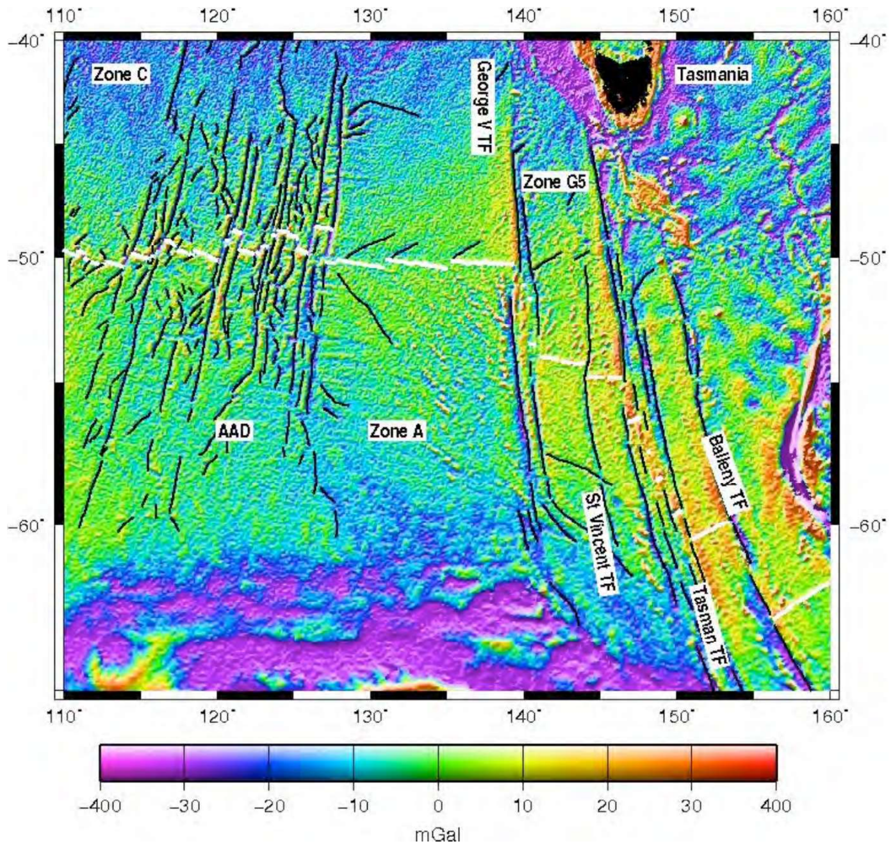


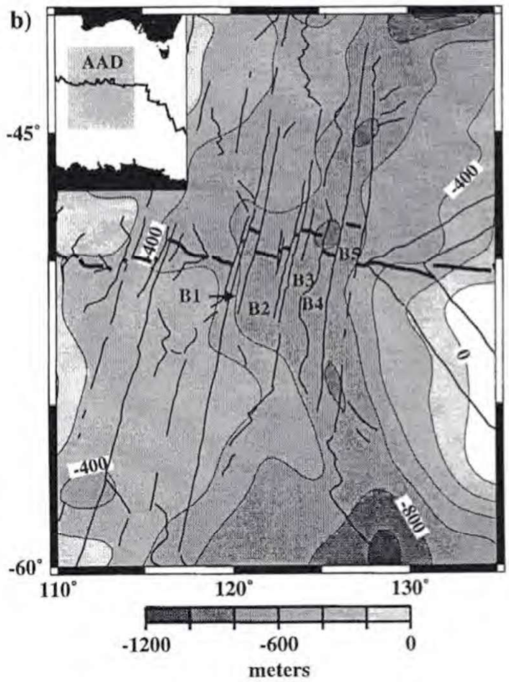
Figure 1: Left : Location of studied area. AMS : Amsterdam-St Paul, BAL : Balleny, KER : Kerguelen hotspots. **Right** : Shaded map of free-air gravity anomalies of southern Indian Ocean between 110ºE and 160ºE (from Sandwell et al., 2009). Bold white line is interpreted ridge axis, in the AAD from (Sempéré et al., 1997), east of the AAD from the gravity data. Thin black lines are traces of ridge-axis discontinuities : fracture zones, propagator pseudofaults, and traces of non-transform discontinuities. AAD: Australian-Antarctic Discordance.

1.1. The Australian-Antarctic Discordance :

The Australian-Antarctic Discordance (AAD) is the section of the SEIR located between 120°E and 128°E, where the ridge axis has the characteristics of a slow-spreading mid-ocean ridge despite its intermediate full spreading rate of 75 mm/y. There, the ridge axis is anomalously deep (4750 m under sea-level), the axial valley is very broad (15 to 20 km) and has a high relief (1500 m), the topography is very rough, and the axis is offset by many transform faults (TF) (Hayes et al., 1972; Weissel, 1974; Sempéré, 1991; Marks, 1999). These characteristics are usually interpreted to reflect the presence of an anomalously cold mantle and thin crust under the AAD (Cochran et al., 1986; Forsyth et al., 1987; Klein et al., 1987; Kuo et al., 1993; Marks, 1990;

Phipps Morgan et al., 1987; Veevers et al., 1982; Weissel et al., 1974). West of the AAD, the ridge characteristics change gradually to a typical intermediate-spreading ridge (Cochran et al., 1997; Ma et al., 1996; Sempéré et al., 1997; Shah et al., 1998), while to the east, the change in morphology occurs abruptly, close to the 128°E transform fault (Christie et al., 1998; West et al., 1994).

The residual depth anomaly (Figure 2), obtained by removing from the bathymetry a model of lithospheric cooling, is characterized by a negative anomaly (seafloor deeper than predicted) over the AAD, about 1200-1700 km wide, extending on the ridge flanks (Hayes et al., 1988; Marks et al., 1990; 1999 ; Whittaker et al., 2010). This suggests that the source of the depth anomaly has always been located under the axis (Marks et al., 1990). The anomaly has a bow (or "C") shape pointing to the west, indicating a westward migration of the anomaly with respect to the axis at about 15 mm/y since the beginning of the opening (Christie et al., 1998; Marks et al., 1990; Veevers, 1982).



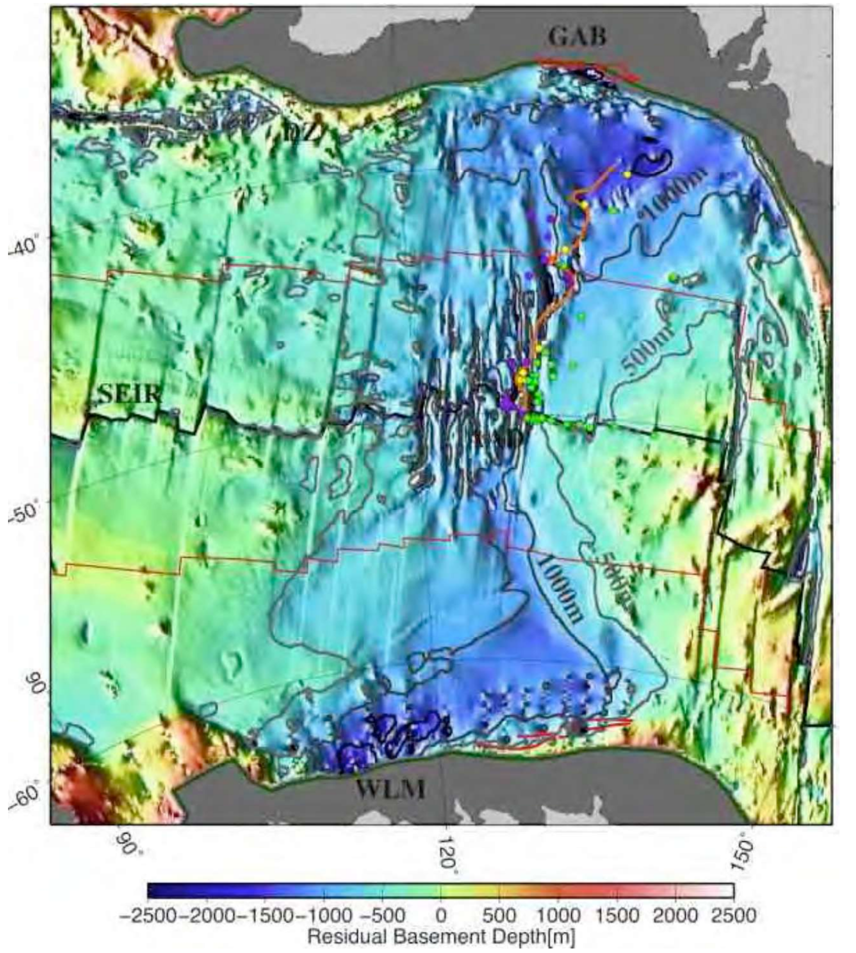


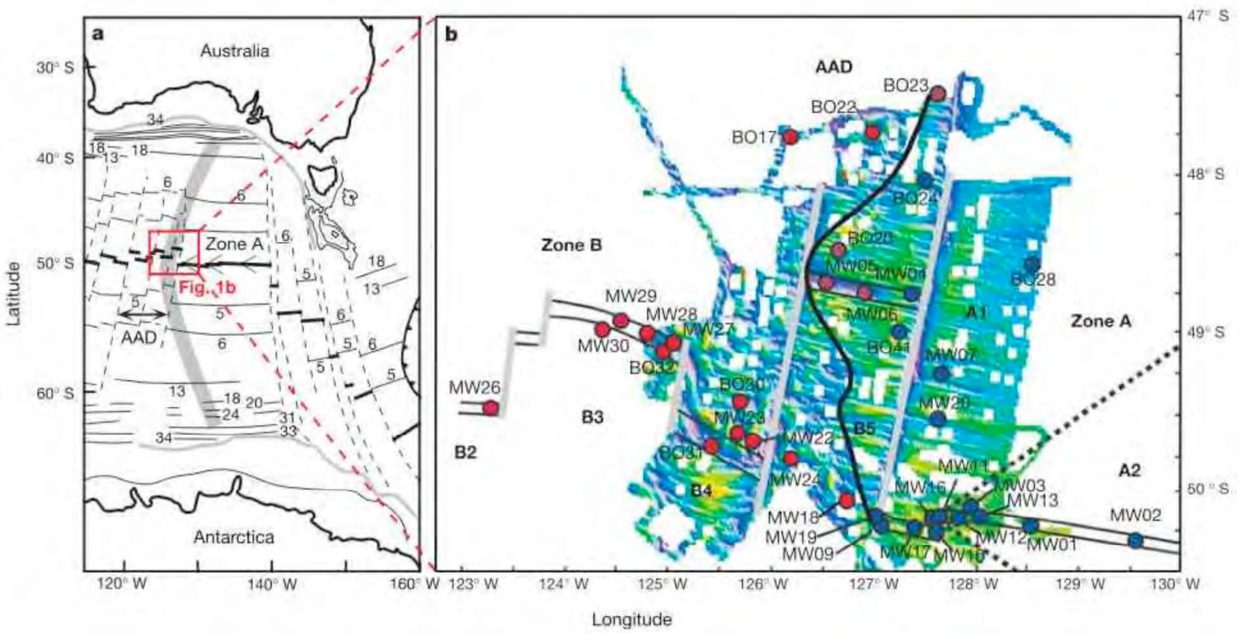
Figure 2 : Left : Depth anomaly in the AAD from Marks et al. (1999). Right : Residual basement depth from Whittaker et al. (2010) (unloaded from the sediments). Basalt geochemistry data (Christie et al., 2004; Kempton et al., 2002) plotted as small circles, green – Pacific-type mantle source, yellow – mixed mantle source, and purple – Indian mantle source.The AAD is marked by a large, deep anomaly extending from margin to margin. The region east of the AAD clearly shows abnormally shallow seafloor.

Weissel and Hayes (1971) and Hayes and Conolly (1972) first described the contrast in morphology and kinematics between the AAD and the adjacent areas, and distinguished three zones from E to W: "zone A" between the 128°E TF and George V TF (140°E), "zone B" in the AAD, and "zone C" west of the AAD (Figure 1). We call "zone G5" the region located between George V TF and Tasman TF (near 146°E). The George V TF is the first in a series of large-offset, right-stepping transform faults. The George V TF has an offset of about 300 km; the Tasman TF (148°E) an offset of about 600 km, and the Balleny TF (155°E) an offset of about 300 km. All these transform faults are multiple-offset transforms, with intra-transform ridge segments.

The SEIR axis morphology varies considerably along the axis, although its full opening rate is intermediate along the whole ridge, varying between 59 and 75 mm/y (Weissel et al., 1971; 1974; Ma et al., 1996; Sempéré et al., 1997; Shah et al., 1998). Between the RTJ and Saint Paul and Amsterdam Islands (88°E), the axis has a slow-spreading ridge morphology, with an axial valley (Royer et al., 1985). From Amsterdam to the 102°45'E TF, and between 128°E and the MTJ, the SEIR axis has a dome-like morphology, as is usually observed on fast-spreading ridges (Ma et al., 1996). Between 102°45'E and the western limit of the AAD (120°E), the axis alternates between a slow-spreading and a fast-spreading morphology (Cochran et al., 1997; Sempéré et al., 1997; Shah et al., 1998). In the AAD, the SEIR has a deep axial valley, as slow-spreading ridges away from hotspots. The axial region in the westernmost part of zone A was surveyed by multibeam bathymetry, showing that the area has a low relief (200 to 400 m variations) and a domed axis, 10 km wide and 500 m high, typical of fast-spreading ridges like the EPR (Sempéré et al., 1996; West et al., 1994).

1.2. The isotopic boundary between Indian and Pacific Mantles

The eastern boundary of the AAD coincides with a major isotopic geochemical boundary between mid-ocean ridge basalts (MORBs) derived from Indian mantle and MORBs derived from Pacific mantle (Figure 3) (Kempton et al., 2002 ; Klein et al., 1988; Pyle et al., 1992). Indian-type MORB has characteristically higher $^{87}Sr/^{86}Sr$, lower $^{143}Nd/^{144}Nd$, and higher $^{208}Pb/^{204}Pb$, $^{207}Pb/^{204}Pb$ ratio at a given $^{206}Pb/^{204}Pb$ value, relative to Pacific and Atlantic MORB (e.g., Dupré and Allègre, 1983 ; Hart, 1984 ; Dosso et al, 1988). Samples from the AAD area include basalts collected on the ridge axis near the eastern boundary of the AAD, and off-axis sampling by ODP drilling (cruises MW88 of R/V Moana Wave, BMRG05 of R/V Melville, DSDP sites 263–267, 278–280 and 282, ODP leg 187, see distribution on Figures 2 and 3 and in document 2). Off-axis sampling lead several authors (Lanyon et al., 1995; Pyle et al., 1995) to infer that the compositional boundary has been migrating relatively fast (25 to 40 mm/y) towards the west in the last 30 m.y. Isotopic compositions of older samples collected by ODP off-axis drilling, however, suggest that the geochemical boundary has not migrated faster than the depth anomaly (Christie et al., 2001; Kempton et al., 2002; Pedersen et al., 2001). Pb isotope data (Klein et al., 1988 ; Pyle et al., 1992 ; Lanyon et al., 1995) map a gradient in $^{206}Pb/^{204}Pb$ that extends from the AAD out to about 140°E, where the data coverage ends, that ranges from relatively low values of the AAD to higher values that approach the Pb isotope signature of Balleny Island basalts.



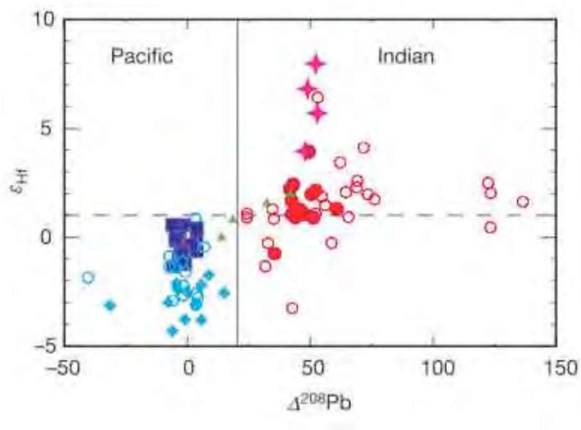
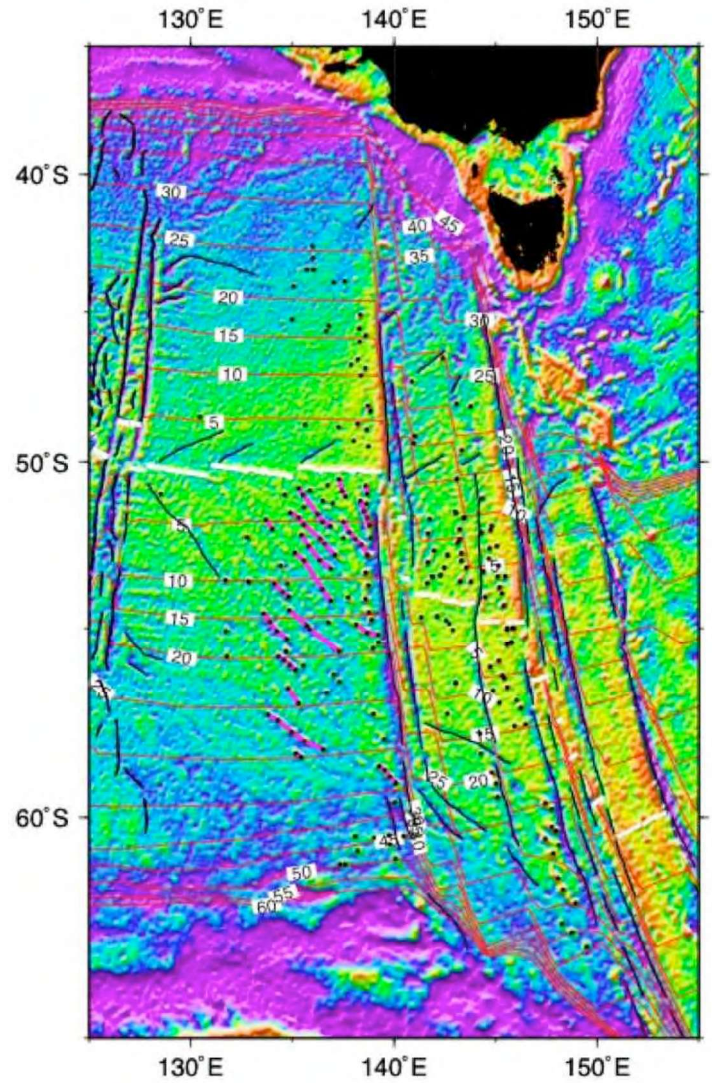


Figure 3: Top. a. General setting of SEIR south of Australia (from Hanan et al., 2004). SEIR axis is shown as thick black lines, main fracture zones as dashed lines, thin black lines are selected magnetic anomalies. Thick grey line: center of the residual depth anomaly. b. Location of the samples in Hanan et al., 2004, on a bathymetry map. Blue circles Pacific type, red circles Indian type, cross-hached circles transition type mantle sources. **Right.** Contrasting $\Delta^{208}Pb$ and $\Delta\varepsilon_{Hf}$ isotopic variations for MORB from the AAD and the Indian Ocean basin in general, versus the Pacific ocean basin from Hanan et al. (2004).

1.3. The anomalous area East of the AAD

The satellite-derived free-air gravity anomaly map (Sandwell et al., 1997, 2009) reveals a zone near the South East Indian Ridge (SEIR), between 128°E and 150°E, where we observe many free-air gravity anomaly highs (Figure 4) (Gomez, 2001 ; Briais et al., 2009). These anomalies are sometimes aligned to form ridges. They contrast with the small roughness of the average gravity anomalies of the seafloor in this zone (Whittaker et al., 2010).

Many of these off-axis seamounts are seen on both ridge flanks, in zones A and G5, but their distribution and signature are not uniform. Many gravity highs form chains in zone A, particularly in its eastern half, except for a few anomalies close to the George V TF and located between 54°S and 55°S which appear isolated. The obliquity of the orientations of the lineations in the southern flank of zone A implies that these constructions do not have their origin in the tectonics of the transform fault. In zone G5, we observe many isolated positive gravity anomalies. Some anomalies close to the 144°E discontinuity or to the George V TF form short lineations.



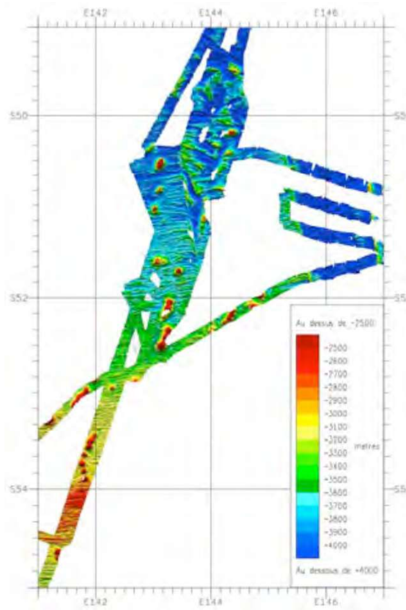


Figure 4: Off-axis seamounts (black dots) and seamount chains (purple lines) identified from the gravity map shown on shaded satellite free-air gravity anomaly map (Gomez, 2001 ; Briais et al., 2009). Black lines mark the traces of the outer pseudo-faults. Red lines are isochrons at 5 Ma interval (Müller et al., 1997). Right: Bathymetry map of SEIR ridge flank from 142°E to 148°E, 50-54°S, showing that the gravity highs correspond to seamounts. From a compilation of multibeam data available in databases (cruises West 11 and Sojourn 6 of R/V Melville).

All volcanoes observed on satellite-derived gravity in the studied zone are located off-axis, on either side of the ridge. The free-air anomaly (FAA) gravity maxima are often surrounded by a minimum of lower amplitude. This minimum probably marks a lithospheric flexure caused by seamount load, suggesting that they were emplaced on young lithosphere but not very close to the axis. Moreover, volcanoes on one flank are not associated to a conjugate structure on the other flank, neither in zone A nor in zone G5, which confirms that those edifices were not built at the axis. No volcano large enough to be identified on altimetry data has been observed at the axis, but on a lithosphere at least 0.5 Ma old in zone G5 and 1 Ma old in zone A (Figure 4). This observation is analogous to the one made in the East Pacific Rise (EPR), where volcanoes identified on bathymetry maps are mostly observed on crust at least 0.1 m.y. old (Alexander et al., 1996). The difference between the SEIR and the EPR in the minimal age of seamount emplacement may reflect the fact that in the EPR, the observations have been made from multibeam bathymetric maps, allowing to identify smaller seamounts closer to the axis.

In zone A, seamount population is much more numerous south of the axis ridge than north of it (Gomez, 2001 ; Briais et al., 2009). In zone G5 the distribution is more balanced on both flanks. However, there are more seamounts north of the axis for low crustal ages. On the southern flank of zone G5, the distribution is also inhomogeneous. The corridor between George V FZ and the 144°E FZ has much less seamounts than the corridor between the 144°E FZ and the Tasman FZ. It is interesting to note that all volcanic chains observed in zone A are located east of 133°40’E, whatever the age of the underlying lithosphere. Moreover, lineations located on a crust younger than 5 Ma are all located east of 135°E, in a section which corresponds to the easternmost segment of the ridge in zone A.

Both regions (A and G5) show a clear maximum in the number of volcanoes versus latitude, between 50 and 56°S for zone A and between 52 and 58°S for zone G5. The 2° offset in the maxima latitudes is much smaller than the offset across the George V TF (about 400 km). There is a minimum in the number of seamounts between 40 and 48°S in both areas, but especially in G5 region. Histograms of the seamount distribution as a function of the crustal age under the seamount reveal some similarities between zones A and G5. In both areas, seamounts are observed on 30 m.y.-old lithosphere in the southern flank of the axis, but not in the northern flank, where they appear only at 26 Ma. All zones have some volcanoes built on lithosphere aged around 24 Ma. In the southern flank of zone A, a continuous increase in the number of volcano starts at this crustal age and shows a maximum between 16 and 18 Ma. In the northern flank of zone A, there is also a maximum between 16 and 18 Ma. The number of volcanoes increases again around 10 Ma, especially in the southern flank of zone A and in the northern flank of zone G5. As noted previously, the minimum crustal age under the seamounts is about 2 Ma.

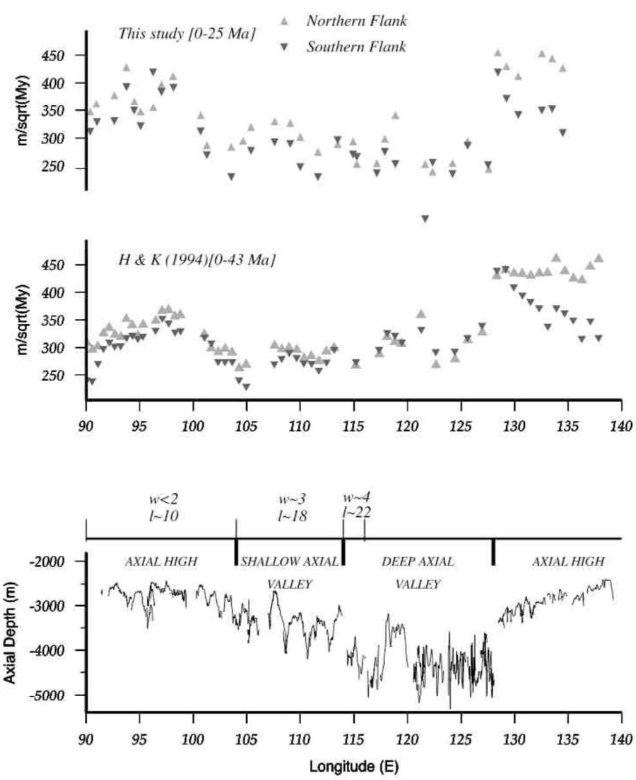


Figure 5: (bottom) Bathymetric profile (plotted versus longitude) along the axis of the South Indian Ridge. Indications on axial morphology are after Cochran et al. [1997]. Average characteristic width (w) and length (l) of abyssal hills are in km, based on work by Goff et al. [1997]. (top) Along-axis subsidence rates variations plotted versus longitude. (middle) The [0 – 42 Ma] age range [after Hayes and Kane, 1994]. (top) This study subsidence rates, reassessed for the 0 – 25 Ma age range. Triangles and inverted triangles are for the northern and southern flanks, respectively. From Géli et al. (2007).

All these observations show that the area East of the AAD is probably as anomalous as the AAD itself. The shallow residual basement depth (Figure 2), off-axis volcanism, and the presence of westward propagators in zone A suggest that the magma production under the South Tasmania area is not only larger than the cold AAD area, but also abnormally high compared to other sections of intermediate-spreading mid-ocean ridges. Other observations support this hypothesis. For example, magnetic anomaly amplitudes are anomalously high in zone A, anomalously weak in the AAD and normal west of the discordance (Anderson et al., 1980; Marks et al., 1990; Vogt and Johnson, 1979; Weissel et al., 1974). A major difference between the AAD and the area east of it is the asymmetry of the distribution of the various anomalies. Whereas the depth, and subsidence anomalies are relatively symmetrically distributed in the AAD with respect to the ridge axis, the area east of the AAD shows strong anomalies on the southern flank of the ridge in zone A, and more off-axis volcanoes on the northern flank in zone G5. The subsidence estimate also shows a strong asymmetry between the two flanks in zone A (Figure 5) (Géli et al., 2002).

Various models have been proposed for the origin of the AAD geophysical anomaly and geochemical boundary. Marks et al. (1990) suggest that the anomalous depth and low melt supply reflects low mantle upwelling or mantle downwelling. Gurnis et al. (1998; 2003) and Ritzwoller et al. (2003) suggest that the AAD geochemical and geophysical characteristics result from the lithospheric slab which subducted beneath the Gondwana in the Mesozoic and is currently drawn up by the SEIR. Lin et al. (2002) point out, however, that drawing up the cold subducted slab would require unrealistic asthenospheric viscosities. Whittaker et al. (2010) suggest that depth and roughness anomalies are due to negative dynamic topography due to the subducted slab, and to crustal thickness variations. West (1997) suggests that a migrating coldspot could result form a trapped fragment of sub-continental mantle beneath the SEIR. Gaina et al. (2000) show that hotspot tracks in the Tasman sea are best modeled using a Pacific hotspot reference frame, and suggest that volcanism, as well as spreading asymmetries in the Tasman sea imply that the region south of Tasmania represents the interaction with a paleo-superswell (e.g., McNutt, 1998). Mantle structure and dynamics beneath the AAD and east of it, at the boundary between Indian and Pacific domains, remain controversial, as the pictures provided by various observations suggest distinct patterns of mantle flow. The STORM project aims to constrain the crustal structure of the SEIR and the mantle flow between the Pacific and the Indian domains.

2. Objectives of the cruise

The STORM cruise has three main objectives:

- to constrain the mantle flow from the Pacific to the Indian domains
- to understand the origin and the emplacement of off-axis volcanic seamounts and seamount chains
- to explore a remote part of the mid-ocean ridge system for hydrothermal activity in various tectonic contexts

These objectives span various geological scales in time and space. The STORM project is a multidisciplinary cruise that will collect data and samples to tackle these problems.

2.1. Mantle circulation beneath the South Tasman area.

Two major observations in the distribution of the off-axis seamounts and chains provide insights into the mantle dynamics in the region. First, most of the identified seamounts are confined to the zone attributed to the isotopic Pacific-type mantle. Second, the volcanic lineations are oblique to spreading, and “point” to the west. These two observations support the hypothesis of asthenospheric flow from east to west under the study area. The distribution in time also suggests that the building of the volcanoes may coincide with the arrival in the area of Pacific-type mantle following the Tasman Rise breakup.

The recent development in the resolution of seismic tomographic models has been used to discuss the existence of a cold lithospheric slab beneath the AAD (Gurnis et al., 2003; Ritzwoller et al., 2003). But the old and the recent tomographic images also suggest the presence of anomalously hot upper mantle east of the George V FZ. A study around Dumont d’Urville shows an 8% decrease of Rayleigh wave group velocities under the region northeast of the station, that is east of the George V fracture zone (140°E) (Rouland et al., 1985). Anomalously slow Rayleigh waves have also been estimated by Roult et al. (1994) and by Danesi and Morelli (2000) in an area centered on 150°E, 65°S, at a depth of about 80–100 km. Estimates of Rayleigh wave group velocities by Ritzwoller et al. (2003) show an anomaly more centered on the eastern axial segment in zone A. The anomaly around 150°E, 65°S coincides with a positive geoid anomaly (Cazenave et al., 1992; Marks et al., 1990). All these images suggest that not only the mantle under the AAD is anomalously cold, but also that the mantle east of it is anomalously hot. These observations have been interpreted to reflect the presence of an immature or aborted hotspot named the “George V Complex” (e.g., Marks et al., 1990). We infer that such a mantle thermal anomaly is likely to be responsible for the off-axis volcanism in zones A and G5.

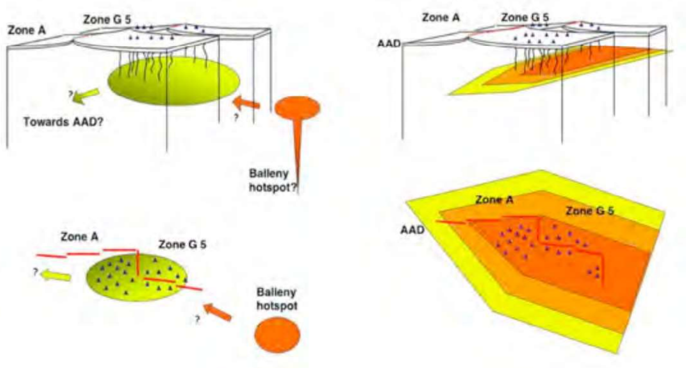


Figure 6: Sketches illustrating possible sources of extra volcanism in zones A and G5, in cross-section (top) and map view (bottom). Left: Broad thermal or chemical anomaly centered near the axis in zone G5 and extending under the southern flank in zone A. Arrows indicate the hypothetical direction of mantle flow between Balleny hotspot and the anomaly, and between the anomaly and the AAD. **Right**: Westward flow of Pacific mantle (marked by the large arrow shape). The isotopic boundary between Pacific-type and Indian-type mantles correspond to the yellow zone. A high melting rate is inferred in the dark orange area, beneath the off-axis volcanoes.

Several hypotheses suggest that asthenosphere flows under the SEIR towards the AAD, that is flowing westward in zone A and eastward west of the AAD (e.g. Forsyth et al., 1987; Marks et al., 1990; Vogt, 1975). Such longitudinal asthenospheric fluxes account for the axial depth anomalies, and for the presence of propagators in zone A and west of the AAD. According to these models, asthenospheric flow would come from far-away hotspots located east and west of the AAD, and would converge at the AAD (Phipps Morgan et al., 1994; Vogt, 1983). Mantle sources would be the Amsterdam and Kerguelen hotspots to the west (Morgan, 1978; Vogt, 1975) and the Tasmantid and Balleny plumes to the east (West et al., 1997). Other models invoke the presence of the ill-defined hotspot, the "George V Complex" under the area (e.g., Marks et al., 1990). The passive separation of Australia and Antarctica continents is also inferred to have resulted in the presence of a colder mantle under the AAD (Lin, 2002), which may imply a significant component of lateral asthenospheric flux (West et al., 1997).

The STORM cruise will test the hypothesis that the observed distribution and geometry of the off-axis volcanism result from the partial melting of asthenospheric Pacific-type mantle as it flows from east to west relative to the lithosphere created at the SEIR. This process likely started with the opening of the oceanic spreading center between Tasmania and Antarctica. The mantle flow at the origin of the magmatic anomaly appears to be partly dammed by the large transform faults, as revealed by the variations of the residual mantle Bouguer gravity anomalies across the fracture zones (Briais et al., 2009). Small-scale convection might accompany the main regional westward flow (Dubuffet et al., 2000), resulting in the development of parallel volcanic chains. The relative motion between the asthenosphere and the plates in the area might only correspond to the eastward absolute motion of the Antarctic and Australian plates. Such a flow could result from the global-scale mantle flux from the Pacific superplume (Romanowicz et al., 2002).

Specific questions to be addressed by the project:

- Are there variations in the isotopic, geochemical, and/or petrological characteristics within the « Pacific-type » domain south of Tasmania?
- Are there differences between the magmas erupted along the ridge axis and those which form the off-axis seamounts?
- What is the relationship between a mantle flow and the large transform faults south of Tasmania?
- When did the flow from Pacific to Indian started?

2.2. Origin of the off-axis volcanoes

The off-axis volcanoes and seamount chains may result from three main processes. First, they could correspond to propagator pseudofaults. Second, they could be tectonic reliefs resulting from intraplate deformation. Finally, they could be volcanic seamounts and seamount chains. A major objective of the cruise is to understand how they form.

Propagators?

Both zone C and zone A are characterized by propagators converging toward the AAD. Propagators are ridge-axis discontinuities where one ridge segment lengthens and the other retreats (e.g., Hey et al., 1977). The traces of the discontinuity are the propagator pseudofaults, forming V-shaped off-axis traces. The pseudofaults have a higher tectonic signature if the offset between the propagating and retreating axial segments is large (Phipps Morgan and Sandwell, 1994). Phipps Morgan and Sandwell (1994) identified three left-stepping propagators in zone A from their signature in the satellite altimetry data (Figure 1), and proposed that propagators "go down" a regional bathymetric or gravimetric gradient. SEIR propagators east or west of the discordance converge towards the AAD, which is the gravimetric minimum and depth maximum of the region. West et al. (1999) suggest that the topographic gradient alone could not explain the observation of propagators on either side of the AAD, and imply additional asthenospheric flow and crustal variations.

Deformation? Recent changes in plate motion?

Another hypothesis to explain the lineations observed on the southern flank of zone A is intraplate compressional or extensional deformation, magmatic or amagmatic. Diffuse lithospheric deformation can be induced by a change in opening direction which creates tectonic reliefs along faults or volcanic ridges.

A clockwise change in opening direction is documented near 4 m.y. (Vogt et al., 1983) or 6 m.y. ago (Cande and Stock, 2004). Also, the 144°E discontinuity traces and the western traces of Tasman and George V TFs form a pronounced bend around 3 Ma of crustal age. These bends might reflect a clockwise rotation of the plates relative motion. Such a rotation increased the lateral offset at left-stepping propagators, and results in the segment propagation into older lithosphere, increasing the relief across the pseudo-faults (Phipps Morgan et al., 1994) (Figure 7).

Gravity and seismicity suggest that recent extension exists, but appears to be limited to the transform fault zones and propagator tips. The altimetry data show that the northern and southern traces of the George V and Tasman transform faults are offset left-laterally. The seismicity (Figure 8) is concentrated near the transform faults and major discontinuities, and possibly along two ridges located near George V TF and not very oblique to its trace. The gravity lineations which are more oblique to the TF do not show any seismicity, suggesting that they do not result from current intraplate deformation. Both altimetry maps and seismicity distribution suggest that the deformation along the TFs is not accommodated on simple shear zones, but on zones about 50 km wide. Transform zone systems might include several intra-transform segments (Figure 4). As shown in the sketch of

Figure 7, extension, at least in the transform faults, might be explained by the recent clockwise rotations in spreading direction. Note that the resulting extension would have the same direction in zone A and zone G5, in the inside corners of the George V TF. The few lineations observed in zone G5, however, do not have the same orientation as those in zone A (Figure 4).

The gravity lineations away from the transform fault zones are not easily explained by diffuse intraplate deformation. Therefore we favor the hypothesis that the gravity highs correspond to volcanic seamounts and seamount chains. Moreover, the amplitude of the gravity lineations in zone A and of the isolated gravity peaks in zones A and G5 resemble those described for the Rano-Rahi area on the flank of the East Pacific Rise (Scheirer et al., 1996), or the “President Jackson” seamounts near Gorda ridge in the northern Pacific (Clague et al., 2000). Mapping of the volcanic ridges and dating of the structures would be necessary to assess the role of tectonism and volcanism in the construction of the gravity lineations.

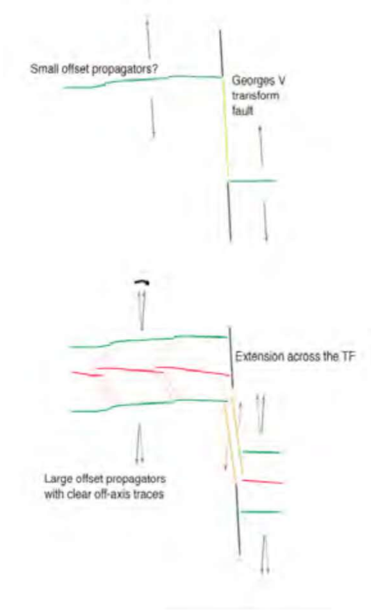
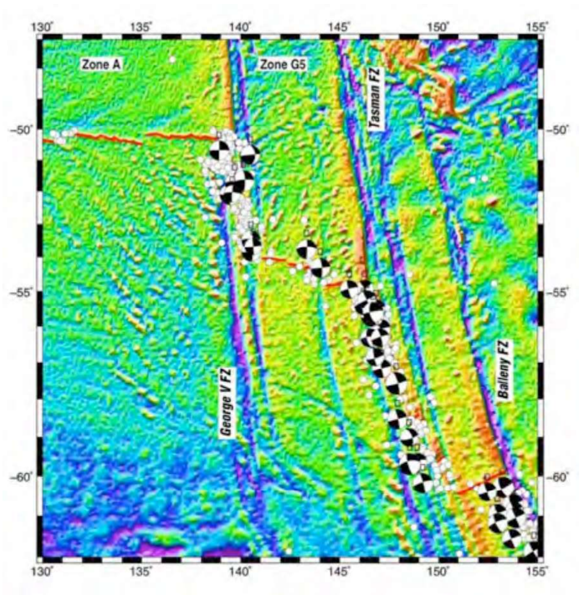


Figure 7: Sketch showing the effects of a clockwise rotation of the opening direction in zone A. The rotation tends to increase the offset of left-stepping propagators, which thus leave clear traces in the seafloor. The rotation is also likely responsible for extension in the George V transform fault.

Figure 8: Epicentres (white dots) of earthquakes of magnitude 1.5 and above recorded from 1973 to 2001 located on a shaded gravity map centered on the transform faults [Harward data center, 2001]. Seismicity is mostly located along transform faults. All earthquake focal mechanisms (black and white circles, lower hemisphere, same source) show left-lateral strike-slip motion.



Another hypothesis to explain off-axis volcanic chain formation is that volcanic chains are created from the building of a first volcano following the Hieronymous and Bercovici (2000) model. This model allows the creation of several parallel chains from the initial lithospheric load of a single volcano, in regions where the local magmatic budget is particularly high, such as in the Rano-Rahi zone near the EPR or the Pacific Superswell. The flexure resulting from the initial load creates a pattern of stresses that form fissures through which other volcanoes are built, and so on until finally a whole chain is formed. In zone A, four lineations oriented about N310°E are relatively parallel to each other. They are located south of the 135°E propagator and on a 2 to 14 Ma aged crust (50°S to 55°S). The distance between chains is about 35 km along isochrons, which is compatible with the model. The orientation of the chains in the model, however, is perpendicular to local minimum stress. In our study area, the obliquity of the volcanic chains and their opposite direction in zone A and zone G5 would require a complex distribution of extensional stresses to apply the model.

Mantle anomalies ?

Yet another hypothesis is the existence of individual magma sources near the axis beneath the seamounts and chains, which cross the young and thin lithosphere to build volcanoes. Such magma sources may be chemical anomalies embedded in the asthenospheric mantle, or thermal mantle anomalies reflecting small-scale mantle convection. The direction of volcanic lineations in both A and G5 regions, with ridges pointing west in both cases, suggest that such mantle sources would migrate at rates ranging between about 25 mm/y to 50 mm/y with respect to the overlaying lithosphere. The lower value coincides with the rate of migration of the isotopic boundary between Pacific and Indian mantle (Christie et al., 1998; Pyle et al., 1995). The higher value coincides with the rate of the propagators.

Specific questions addressed by the project:

- What is the relation between the mantle sources of the ridge basalts and those of the seamounts?
- How deep is the mantle that melts to form off-axis seamounts?
- What is the relation between volcanic chains and propagators?

- Are there signs of recent deformation in the area?

2.3. Exploration for hydrothermal activity:

A variety of tectonic contexts are present in the study area: the AAD cold and rough area to the west, robust mid-ocean ridge segments in zone A, recent off-axis volcanoes, some of them might still be active, large transform fault system at 140°E (George V transform fault), probably including intra-transform segments, and transform ridges. Deformation related to the transform faults and propagator tips as suggested from gravimetry and seismicity, can favour fluid circulation within the seafloor. This process is evidenced by the degree of alteration of rocks whereby the mineralogical assemblage observed in them is a direct record of the nature of the protolith and its interaction with the percolating fluid. The idea is to get a comprehensive view into the links between deformation, rock alteration and fluid circulation, which can be assessed by a combined approach of seafloor mapping, rock sampling and hydrothermal fluid detection.

Because of its remoteness and the high seas, the area has not been surveyed yet for hydrothermalism. One aim of the project is to detect signals characteristic of hydrothermal activity in the water column, which would allow us to constrain on a local level new hydrothermal vent sites in such a remote area at the boundary between Indian and Pacific oceans. Even if AUV or ROVs may not be launched in the expected high seas in the study area, an exploratory approach using new chemical in-situ technologies would help us in identifying potential hydrothermal discharge sites. Hydrothermal activity can be expressed as high (up to 400ºC) or low (<100ºC) temperature hydrothermal fluid expelled from the seafloor either on the ridge axis or off-axis as observed on the mid-Atlantic ridge for example (e.g., Melchert et al., 2008). The key resides in acquiring on board chemical/physical features in the water column attesting of hydrothermal activity such as temperature, light backscattering, methane concentration, and total dissolved manganese (TDM). Indeed, a methane enrichment of the water column compared with the background (0.5 nM), associated with low manganese concentrations and no significant turbidity (nephelometry) signal, provides evidence of fluid circulation into ultramafic rocks (Charlou et al., 1991). The discovery of the off-axis ultramafic-hosted Lost City vent site on the Atlantis massif core complex (mid-Atlantic ridge), showed evidence for low temperature hydrothermal venting, producing low Mn/CH_4 and low nephelometry/CH_4 signals (Kelley et al., 2001). When high temperature hydrothermal fluis are present, we usually observed a significant turbidity signal associated with high TDM/CH_4 ratio (Charlou et al., 1991).

To maximise the chances of intercepting non-buoyant hydrothermal plumes, we propose a combined approach measuring optical backscatter to detect suspended particulate matter in the overlying water column — diagnostic of dispersing non-buoyant hydrothermal plumes-, temperature anomalies, and in-situ chemical sensors (Mn and CH_4).

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